INCH-POUND

MIL-B-40098H
26 June 1989
SUPERSEDING
MIL-B-40098G
30 March 1982

MILITARY SPECIFICATION

BURNER UNIT, GASOLINE, FIELD RANGE OUTFIT, M2A

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a burner unit designed for use with leaded or unleaded gasoline (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

FEDERAL

QQ-N-290	- Nickel Plating (Electrodeposited)
PPP-B-601	- Boxes, Wood, Cleated-Plywood
PPP-B-636	- Boxes, Shipping, Fiberboard

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: U.S. Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A FSC 7310

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MILITARY

MIL-P-116	- Preservation, Methods Of
MIL-B-7883	- Brazing of Steels, Copper, Copper Alloys, Nickel Alloys, Aluminum and Aluminum Alloys
MIL-S-7916	- Sealing Compound, Thread and Gasket, Fuel, Oil, and Water Resistant
MIL-C-40077	- Cabinet, Gasoline Field Range Outfit, Model 59, and Cradle, Cook Pot

STANDARDS

FEDERAL

FED-STD-H28	- Screw-Thread Standards for Federal Services

MILITARY

MIL-STD-105	- Sampling Procedures and Tables for Inspection by Attributes
MIL-STD-129	- Marking for Shipment and Storage
MIL-STD-130	- Identification Marking of US Military Property
MIL-STD-147	- Palletized Unit Loads
MIL-STD-210	- Climatic Information to Determine Design and Test Requirements for Military Systems and Equipment

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Naval Publications and Forms Center, (ATTN: NPODS), 5801 Tabor Avenue, Philadelphia, PA 19120-5099.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS

U.S. ARMY NATICK RESEARCH, DEVELOPMENT, AND ENGINEERING CENTER

5-11-1225	- Burner Unit, Gasoline, Field Range Outfit, M2A; Assembly Complete
5-11-1226	- Frame Assembly
5-11-1227	- U Tank Assembly
5-11-1228	- Bracket Assembly
5-11-1229	- Filler Cap and Air Valve Assembly
5-11-1230	- Preheater Assembly
5-11-1232	- Preheater Generator, Needle, Valve Knob and Fuel Cup Assemblies
5-11-1236	- Air Shutter and Flame Valve Knob Assemblies

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5-11-1237 - Head, Burner
5-11-1238 - Generator Assembly
5-11-1239 - Orifice Nut Assembly
5-11-1240 - Mixing Chamber Assembly
5-11-1242 - Feed Tube Assembly
5-11-1243 - Top Shield and Spring Hook Assembly
5-11-1244 - Preheater Shield Assembly
5-11-1668 - Safety Device Assembly and Details

(Copies of drawings are available from the US Army Natick Research, Development, and Engineering Center, Natick, MA 01760-5014.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DOD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

B 633 - Electrodeposited Coatings of Zinc on Iron and Steel

D 3951 - Standard Practice for Commercial Packaging

E 18 - Rockwell Hardness and Rockwell Superficial Hardness of Metallic Materials

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

(Non-Government standards and other publications are normally available from the organizations that prepare or distribute the documents. These documents also may be available in or through libraries or other informational services.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 6.3) in accordance with 4.3.

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3.2. Materials and components. Materials and components shall be as specified herein and on the applicable drawings. Materials and components not definitely specified shall be of the quality normally used by the manufacturers provided the completed items comply with all provisions of this specification. It is encouraged that recycled material be used when practical as long as it meets the requirements of this specification.

3.3 Design and construction. The design and construction of the burner units shall be as specified herein, and as shown on assembly Drawing 5-11-1225 and detail drawings called out thereon. "Normal operating conditions" shall be defined as the minimum requirement for ground equipment as stated in MIL-STD-210 (see 2.1).

3.3.1 Brazing. All brazed joints and closures shown on the drawings, and specified herein, shall conform to MIL-B-7883. U tank halves, U tank fittings, and generator fittings shall conform to type II, grade B of MIL-B-7883. All other brazing including minor repairs to U tank halves and U tank generator fittings shall conform to type I of MIL-B-7883.

3.3.2 Threads and threaded joints. All threads shall conform to FED-STD-H28.

3.3.2.1 Threaded pipe joints. All threaded pipe joints shall be sealed with a compound conforming to MIL-S-7916.

3.3.3 Gage, air pressure. The air pressure gage shall be 1-1/2 inch size having a drawn steel case and bezel finished with epoxy base black enamel. The crystal shall be a heavy, flat, clear glass. The bourdon tube shall be silver brazed, in conformance with type I of MIL-B-7883, at the socket and end piece. A restrictor screw having an orifice of 0.018 inch plus 0.0005 inch, minus 0.000 inch diameter shall be inserted in the socket.

3.3.4 U-tank assembly. The U-tank assembly shall withstand an internal air pressure of 75 +5/-0 pounds per square inch gage (psig) and shall not leak, distort, rupture or crack when used under normal operating conditions (see 4.5.1).

3.3.5 Generators. After complete fabrication, the generators shall withstand an internal air pressure of 125 +5/-0 psig, and shall not leak, distort, rupture, or crack under normal operating conditions (see 4.5.2).

3.3.6 Burner unit. The assembled burner unit shall withstand an internal air pressure of 25 +5/-0 psig and shall not leak under normal operating conditions (see 4.5.3).

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3.3.7 Alignment.

3.3.7.1 Generator assembly. The generator assembly shall conform to Drawing 5-11-1238. The generator assembly shall fit into the burner unit without application of any distortion force to either the generator assembly or the burner unit (see 4.5.4.1).

3.3.7.2 Burner unit. The burner unit frame assembly shall conform to Drawing 5-11-1226. A complete generator assembly shall fit into location within the burner unit without application of any distortion forces to either the burner unit or the generator assembly (see 4.5.4.2).

3.3.7.2.1 Burner unit allignment. After complete assembly, the burner unit shall be square within 1/16 inch when measured at any point around the frame body to ensure a proper fit into the cabinet for a Model 59 Gasoline Field Range.

3.4 Performance. The assembled burner units shall be tested for performance as specified in 4.5.5 at an ambient temperature of 70 ± 10°F. The preheater shall produce sufficient heat to allow the flame valve to be opened and the burner ignited from the preheater flame within a 3-minute starting period. During preheating, the pressure shall not drop more than 5 psig. Proper flame adjustment shall also be obtained in accordance with 4.5.5.

3.4.1 Safety device assembly. The safety device assembly valve shall release the tank pressure at 60 ± 5 psig and shall reset and hold the pressure at 35 ± 10 psig (see 4.5.6).

3.5 Finish. All finishes shall be as shown on the drawings and as specified herein. Corrosion-resisting steel parts shall be clean and free of dirt, scale, embedded particles, rust stains, rust pits, and other foreign matter. Appropriate treatments may be given, during or after fabrication, to obtain this surface condition.

3.5.1 Enamel finish. Surfaces to be enameled shall be clean and given one coat of gasoline-resistant, aluminum color enamel.

3.5.2 Plating.

3.5.2.1 Zinc coating. Except for fasteners, electrodeposited plating shall conform to type I, classification Fe/Zn5 Sc1 of ASTM B 633. Finish for fasteners shall be as specified in 3.2.

3.5.2.2 Nickel plating. Electrodeposited nickel plating shall conform to class 1, grade F of QQ-N-290.

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3.6 Marking.

3.6.1 Instruction marking. Decalcomanias containing operating and maintenance instructions as specified herein shall be affixed to the right and left of the U-tank as shown on the Drawing 5-11-1227. The words, "BURNER UNIT, GASOLINE, M2A," shall be in bold type at the top of the instructions. Decalcomania with caution instructions shall be as shown on the Drawings 5-11-1225 and 5-11-1242.

3.6.1.1 Operational instructions. The following operational instructions shall be in upper case letters except that "OPERATING INSTRUCTIONS," "SAFE," and "CAUTION - DO NOT RELEASE PRESSURE NEAR OPEN FLAME" shall be in bold type.

OPERATING INSTRUCTIONS

1. CLOSE PREHEATER VALVE AND GENERATOR VALVE BY TURNING CLOCKWISE.

2. STAND BURNER UNIT IN VERTICAL POSITION, REMOVE FILLER CAP, AND FILL WITH 2-1/8 GALLONS OF GASOLINE (LEADED OR NONLEADED). DO NOT OVERFILL. REPLACE AND TIGHTEN FILLER CAP, FINGER TIGHT ONLY. REMOVE PREHEATER SHIELD FROM BOTTOM OF UNIT.

3. PLACE BURNER UNIT IN HORIZONTAL POSITION, REMOVE AIR VALVE CAP, AND ATTACH HAND PUMP.

4. PUMP BURNER UNIT UNTIL AIR PRESSURE GAGE NEEDLE INDICATES 6 TO 8 POUNDS PRESSURE, DISCONNECT PUMP, REPLACE AND TIGHTEN VALVE CAP.

5. IN AN OPEN AREA AWAY FROM FUEL OR KITCHENS, CLEAN PREHEATER ORIFICE BY ROTATING ORIFICE CLEANER CONTROL, LEAVE CONTROL IN DOWN POSITION, PLACE LIGHTED MATCH TO PREHEATER HEAD, TURN PREHEATER VALVE KNOB ONE-QUARTER TURN COUNTERCLOCKWISE, AND IGNITE. ALLOW TO BURN ABOUT 5 TO 7 SECONDS OR UNTIL THE FLAME BURNS EVENLY. OPEN PREHEATER VALVE BY TURNING VALVE KNOB FULLY COUNTERCLOCKWISE. PLACE PREHEATER SHIELD OVER PREHEATER AND GENERATOR. UNDER COLD WEATHER CONDITIONS THE USE OF A SECOND PREHEATER SHIELD, IF AVAILABLE, WILL SHORTEN PREHEATING TIME.

6. PREHEAT UNTIL THE FULL LENGTH OF THE GENERATOR TUBE IS HOT TO THE TOUCH.

7. OPEN FLAME VALVE COUNTERCLOCKWISE SLOWLY UNTIL FLAME IS ESTABLISHED. BURNER SHOULD IGNITE BEFORE VALVE IS FULLY OPEN.

8. REMOVE PREHEATER SHIELD AND ADJUST AIR SHUTTER TO MAINTAIN A SEA GREEN COLOR. TURN PREHEATER VALVE OFF. IF PULSING SOUND IS PRESENT, PARTIALLY CLOSE FLAME VALVE UNTIL A STEADY SOUND IS HEARD AND FIRE IS BURNING EVENLY.

9. BURN FOR 15 MINUTES, THEN MOVE TO KITCHEN.

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10. ADJUST FLAME TO DESIRED HEIGHT, MAKING SURE IT IS CONTAINED DIRECTLY UNDER THE COOKING UTENSIL.

11. MAINTAIN OPERATING PRESSURE WITHIN "SAFE" OPERATING RANGE ON AIR PRESSURE GAGE DIAL.

12. IF PRESSURE ON THE GAGE READS BETWEEN 20 TO 25 PSIG, TURN FLAME DOWN TO ABOUT 1/2 SIZE AND CONTINUE OPERATION FOR 1/2 HOUR. IF PRESSURE PERSISTS BEYOND 25 PSI, TURN UNIT OFF, REMOVE FROM CABINET, AND ALLOW TO COOL BEFORE PLACING BACK IN OPERATION.

13. TO SHUT OFF BURNER UNIT, TURN FLAME VALVE KNOB CLOCKWISE UNTIL TIGHT. ALLOW BURNER UNIT TO COOL. AWAY FROM KITCHEN, RELEASE PRESSURE BY UPENDING UNIT AND SLOWLY LOOSEN FILLER CAP. CAUTION: DO NOT RELEASE PRESSURE NEAR OPEN FLAME.

3.6.1.2 Maintenance instruction. The following maintenance instruction shall be in upper case letters except that "MAINTENANCE INSTRUCTIONS" and "CAUTION: DO NOT TIGHTEN JOINTS WHILE UNIT IS UNDER PRESSURE" shall be in bold type.

MAINTENANCE INSTRUCTIONS

GOOD MAINTENANCE MEANS GOOD OPERATION.
HAVE AT LEAST ONE SPARE GENERATOR ATTACHED.
MAINTAIN A PRESSURE TIGHT UNIT BY PERIODICALLY CHECKING CONNECTIONS FOR TIGHTNESS.

CAUTION: DO NOT TIGHTEN JOINTS WHILE UNIT IS UNDER PRESSURE.

KEEP BURNER SLOTS CLEAN WITH BURNER SLOT CLEANER BLADE PROVIDED.

WHEN PRESSURIZED MIXTURE FROM PREHEATER VALVE FAILS TO BURN CONTINUOUSLY DURING PREHEATING, PERFORM ONE OR MORE OF THE FOLLOWING STEPS:

1. CLEAN PREHEATER ORIFICE BY TURNING ORIFICE CLEANER HANDLE SEVERAL TIMES AND LEAVE HANDLE IN DOWN POSITION AND CHECK PREHEATER BY RELIGHTING.

2. CLOSE PREHEATER VALVE, UNSCREW BURNER HEAD, REMOVE PREHEATER GENERATOR ASSEMBLY, AND CLEAN OR REPLACE GENERATOR SCREEN, CHECK CLEANING NEEDLE; IF BENT OR BROKEN, REPLACE GENERATOR. REPLACE BURNER HEAD AND RELIGHT.

TO CORRECT LOW FLAME ON MAIN BURNER, CLOSE AND OPEN FLAME VALVE SEVERAL TIMES TO CLEAN THE ORIFICE AND RELIGHT.

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IF LOW FLAME PERSISTS ON MAIN BURNER:

1. SHUT OFF UNIT.

2. ALLOW TO COOL.

3. RELEASE TANK PRESSURE.

4. REMOVE TOP SHIELD BY LIFTING CORNER RETAINING SPRINGS CLEAR OF SHIELD.

5. LOOSEN GAS INLET HEXNUT ON MAIN BURNER GENERATOR.

6. DISENGAGE FLAME VALVE END OF MAIN BURNER GENERATOR FROM AIR SHUTTER AND LIFT GENERATOR FROM UNIT.

7. REPLACE WITH NEW MAIN BURNER GENERATOR.

8. REPLACE TOP SHIELD AND FASTEN WITH CORNER RETAINING SPRINGS.

IF BURNER UNIT FAILS TO FUNCTION, TURN THE BURNER UNIT INTO THE SUPPORTING FIELD MAINTENANCE ORGANIZATION.

3.6.2 Identification marking. Identification marking shall conform to the requirements shown on the drawings and in MIL-STD-130. Unless otherwise shown on the drawings, characters shall be 1/8 to 1/4 inch high.

3.7 Workmanship.

3.7.1 Metal fabrication. Metal sheet, strip, rod, and tube used in the fabrication of the burner unit shall be free from kinks, dents, slivers, and sharp edges. Straightening of materials before fabrication shall be done by methods that will not harm the metal. Shearing shall be done neatly and accurately and corners shall be square and true. All screw and rivet holes shall be accurately sized and located, and shall have all burrs removed. All threaded fasteners shall be drawn tight.

3.7.2 Riveting. Rivets shall be tight. The heads shall be neatly finished and in full contact with the surfaces of the mating parts.

3.7.3 Welding, soldering, and brazing. All soldering, welding, and brazing shall be clean, free of flux, pits, and fractures, and shall be continuous, adherent, and not burned through.

3.7.4 Surface finish. All plated and enameled surfaces shall be adherent, not tacky, and free from blisters, flaking, crinkling, runs, embedded foreign material, and areas of no finish.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 Responsibility for dimensional requirements. Unless otherwise specified in the contract or purchase order, the contractor is responsible for ensuring that all specified dimensions have been met. When dimensions cannot be examined on the end item, inspection shall be made at any point or at all points in the manufacturing process necessary to ensure compliance with all dimensional requirements.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).
b. Quality conformance inspection (see 4.4).

4.3 First article inspection. When a first article is required (see 3.1 and 6.2), it shall be examined for the defects specified in 4.4.3 and 4.4.4 and tested as specified in 4.5.3, 4.5.4.1, 4.5.4.2, 4.5.4.3, 4.5.5, and 4.5.6.

4.4 Quality conformance inspection. Unless otherwise specified, sampling for inspection shall be performed in accordance with MIL-STD-105.

4.4.1 Component and material inspection. In accordance with 4.1, components and materials shall be inspected in accordance with all the requirements of referenced documents unless otherwise excluded, amended, modified, or qualified in this specification or applicable purchase document.

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4.4.1.1 Component testing.

4.4.1.1.1 Leak testing. Each U-tank assembly and each generator assembly shall be tested as specified in 4.5.1 and 4.5.2, as applicable, with no evidence of failure. Failure to pass these tests shall be cause for rejection of the defective U-tank assembly or generator assembly, as applicable.

4.4.1.1.2 Hardness testing. The hardness of the flame valve stem and orifice nut insert shall be tested as specified in ASTM E 18 to determine compliance with the requirements specified on Drawing 5-11-1239. The lot size shall be expressed in units of heat-treated flame valve stems or orifice nut inserts, as applicable. The sample unit shall be one flame valve stem or orifice nut insert, as applicable. Three readings shall be taken on each sample unit. The average of the three readings shall be the harness value of the sample unit. The inspection level shall be S-2 (see 6.5).

4.4.2 In-process inspection. Examination shall be made at appropriate stages of manufacture to determine compliance with applicable requirements in table I. Whenever nonconformance is noted, correction shall be made to the applicable process and all items processed. The Government reserves the right to exclude from consideration for acceptance, any material or service for which in-process inspection has indicated nonconformance.

TABLE I. Process examination

Operation	Requirement
Brazing	3.3.1
Sealing of threads and joints	3.3.2
Nickel plating	3.5.2.2
Inert gas shielded arc welding of U-tank (when used as an option)	Dwg 5-11-1227 (View "G")
Heat treatment of generator tube	Dwg 5-11-1232
Zinc plating, air shutter body	Dwg 5-11-1236 and 3.5.2.1
Steel wool, fiber length	Dwg 5-11-1238

4.4.3 End item visual examination. The end items shall be examined for the defects listed in table II. The lot size shall be expressed in units of completely assembled burner units. The sample unit shall be one burner unit. The inspection level shall be II (see 6.5).

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TABLE II. End item visual defects

Examine	Defect	Classification Major	Classification Minor
Finish	Type of finish not as specified	101	
	Blistered, crinkled, not adherent, flakes, or color not as specified	102	
	Permanently stained, tacky, uncoated areas, runs, or embedded foreign matter		201
Design, construction and workmanship, general (applicable to all components)	Any component or assembly missing or not as specified	103	
	Any component or assembly fractured, split, cracked, delaminated, cut, punctured, dented, sprung, warped, creased, malformed, or deteriorated	104	
	Any component or assembly misplaced or not in proper alignment	105	
	Evidence of porosity, cast or forged improperly	106	
	Sharp burrs or slivers	107	
	Functioning component or assembly inoperative	108	
	Loose, not connected or joined as specified	109	
	Not clean (e.g., loose foreign matter in fuel tank)	110	
Rivets, pins, and rivnuts	Not peened or swagged, loose	111	
Bolts, nuts, studs, fittings and all other types of threaded fasteners	Not specified type or thread size, stripped	112	
	Loose		202
Welding and brazing or soldering	Not specified type, missing, fractured, not continuous, not adherent, slag inclusion, scale or flux deposits not removed	113	

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TABLE II. End item visual defects (cont'd)

Examine	Defect	Classification Major	Minor
Decalcomanias	Missing, not specified type, incorrect, incomplete, not legible, not in proper location, or not affixed or made in the prescribed manner	114	
Identification marking	Missing, not specified type, incorrect, incomplete, not legible, not in proper location, or not affixed or made in the prescribed manner		203

4.4.4 End item dimensional examination. The end items shall be examined for conformance to the dimensions specified on the drawings. Only those dimensions that can be evaluated without damaging or disassembling the end items shall be examined. Any dimension not within the specified tolerance shall be classified as a defect. The lot size shall be expressed in units of completely assembled burner units. The sample unit shall be one burner unit. The inspection level shall be S-2 (see 6.5).

4.4.5 End item testing. The end item shall be tested as specified in 4.5.4.1, 4.5.4.2, 4.5.4.3, 4.5.5, and 4.5.6. The lot size shall be expressed in units of completely assembled burner units. The sample unit shall be one burner unit. The inspection level shall be S-4 and the AQL, expressed in terms of defects per hundred units, shall be 4.0.

4.4.5.1 Burner unit leak testing. Every end item shall be tested as specified in 4.5.3. Any test failure shall be cause for rejection of the end item.

4.4.6 Packaging examination. The fully packaged end items shall be examined for the defects listed below. The lot size shall be expressed in units of shipping containers. The sample unit shall be one shipping container fully packaged. The inspection level shall be S-2 (see 6.5).

Examine	Defect
Marking (exterior and interior)	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application
Preservation	Preservation improperly applied or missing

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Examine	Defect
Materials	Any component missing, damaged, or not as specified
Workmanship	Inadequate application of components, such as incomplete sealing or closure of flap, improper taping, loose strapping, or inadequate stapling Bulged or distorted container

4.4.7 Palletization examination. The fully packaged and palletized end items shall be examined for the defects listed below. The lot size shall be expressed in units of palletized unit loads. The sample unit shall be one palletized unit load, fully packaged. The inspection level shall be S-1 (see 6.5).

Examine	Defect
Finished dimensions	Length, width, or height exceeds specified maximum requirement
Palletization	Pallet pattern not as specified Interlocking of loads not as specified Load not bonded
Weight	Exceeds maximum load limits
Marking	Omitted; incorrect; illegible; of improper size, location, sequence, or method of application

4.5 Methods of inspection.

4.5.1 U-tank assembly leak test. Seal all openings in the U-tank assembly except the air inlet. Supply a source of air to the air inlet and apply an air pressure of 75 psig. Immerse the assembly in clean water for not less than 30 seconds. Any evidence of leaks, distortion, rupture, or cracks shall constitute failure of this test to comply with 3.3.4. After testing, release the pressure and thoroughly dry the assembly.

4.5.2 Generator assembly leak test. Close the flame valve and apply an internal air pressure of 125 psig through the inlet end of each generator assembly. Immerse the generator assembly in clean water for not less than 30 seconds. Any evidence of leaks, distortion, rupture, or cracks shall constitute failure of this test to comply with 3.3.5. After completion of the test, release the pressure through the flame valve and thoroughly dry the unit.

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4.5.3 Burner unit leak test. Securely close valves and other openings from the fuel system, except air inlet. Supply a source of air to the air inlet until the air gage of the burner unit indicates an air pressure of 25 psig. Let the burner unit with the applied air pressure stand for a minimum of 15 minutes. Any reduction of air pressure gage reading shall constitute a failure of this test to comply with 3.3.6.

4.5.4 Alignment test.

4.5.4.1 Generator assembly to burner unit. Without applying distortion force, place the generator assembly into a gaging fixture constructed to simulate a completed burner unit (minus the generator assembly). Any force required to ensure a proper fit into the gaging fixture shall constitute failure of this test to comply with 3.3.7.1.

4.5.4.2 Burner unit to generator assembly. Without applying distortion force, place a gaging fixture constructed to simulate a generator assembly in a complete burner unit (minus the generator assembly). Any force required to ensure a proper fit into a gaging fixture shall constitute failure of this test to comply with 3.3.7.2.

4.5.4.3 Burner unit to field range. Place a complete burner unit assembly into a gaging fixture constructed to simulate the metal cabinet specified in MIL-C-40077. Any evidence of out of square alignment in excess of 1/16 inch shall constitute failure of this test to comply with 3.3.7.2.

4.5.5 End item burner unit performance test. Charge the burner unit with sufficient leaded or nonleaded gasoline to conduct the test and 10 to 20 psig of air. Light the preheater so it produces sufficient heat to allow the burner to be ignited from the preheater frame within a 3 minute starting period. The pressure loss during this phase shall not exceed 5 psig. After ignition of the burner, turn off the preheater and operate the unit for approximately 10 minutes over the full range of burning rates at 25 psig (initial repressurization will be permitted). Throughout these burning rates, adjust the air shutter to give a green color flame at the burner surface over a minimum of 75 percent of the burner area. Dissassemble the generator from the first tested unit and use it on all subsequent units given this performance test. Failure to function in accordance with the foregoing shall constitute noncompliance with requirements in 3.4. Upon completion of testing, the generator used in these tests shall be replaced with a new generator. In addition, all gasoline and fumes shall be purged from end items subjected to this performance test.

4.5.6 End item safety device assembly valve test. The end item shall be air pressure tested by filling the fuel tank with pressurized air until the safety valve opens and releases the pressure. Failure of the safety valve to comply with the release and holding pressure values of 3.4.1 shall constitute failure of this test.

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5. PACKAGING

5.1 Preservation. Preservation shall be level A or Commercial, as specified (see 6.2).

5.1.1 Level A. The burner head of each burner unit shall be cleaned process C-1, thoroughly dried, and coated with type P-7 preservative of MIL-P-116.

5.1.2 Commercial. Burner units shall be preserved in accordance with ASTM D 3951.

5.2 Packing. Packing shall be level A, B, or Commercial, as specified (see 6.2).

5.2.1 Level A packing. Each burner unit, preserved as specified in 5.1, shall be packed in a cleated-plywood shipping container conforming to overseas type, style optional of PPP-B-601. Inside dimensions of each shipping container shall be approximately 23-1/4 inches in length, 19-1/4 inches in width, and 9-3/4 inches in depth. Approximate dimensions are furnished as a guide only. An inner packing hold-down device, fabricated from weather-resistant corrugated fiberboard, shall be positioned over the generator assembly at the far end to prevent lateral movement. Each container shall be provided with a pad of the same fiberboard material positioned on top of the burner unit, folded down both sides of the burner to fill void space between the top, side frame members of the burner and the sides of the shipping container. An additional pad of the same fiberboard material shall be positioned at one end of the shipping container to provide physical protection for the pressure gage and other burner controls.

5.2.2 Level B packing. Each burner unit, preserved as specified in 5.1, shall be packed in a fiberboard shipping container conforming to style RSC, type CF (variety SW) or SF, class domestic, grade 275 of PPP-B-636. Inside dimensions of each shipping container shall be approximately 23-1/4 inches in length, 19-1/4 inches in width, and 9-3/4 inches in depth. Approximate dimensions are furnished as a guide only. Each container shall be provided with fiberboard hold-down device and pads as specified in 5.2.1, except the material shall be class domestic, grade 275. Each fiberboard container shall be closed in accordance with method II as specified in the appendix of PPP-B-636.

5.2.2.1 Weather-resistant fiberboard container. When specified (see 6.2), the fiberboard shipping container shall be a grade V3c, V3s, or V4s fiberboard box fabricated in accordance with PPP-B-636 and closed in accordance with method III as specified in the appendix of PPP-B-636.

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5.2.3 Commercial packing. Burner units, preserved as specified in 5.1 and provided with fiberboard hold-down device and pads as specified in 5.2.1, shall be packed in accordance with ASTM D 3951.

5.3 Palletization. When specified (see 6.2), burner units, packed as specified in 5.2, shall be palletized on a 4-way entry pallet in accordance with load type I or Ia, as applicable, of MIL-STD-147. Pallet type shall be type I (4-way entry), type IV, or type V in accordance with MIL-STD-147. Each prepared load shall be bonded with primary and secondary straps in accordance with bonding means K and L or film bonding means O or P. Pallet pattern shall be in accordance with the appendix of MIL-STD-147. Interlocking of loads shall be effected by reversing the pattern of each course.

5.4 Marking. In addition to any special marking required by the contract or purchase order, shipping containers shall be marked in accordance with MIL-STD-129 or ASTM D 3951, as applicable.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The burner unit is intended to be used with Cabinet, Gasoline Field Range Outfit Model 59 and Cradle, Cook Pot; Oven, Baking and Roasting, Deck, Gasoline, 3 Compartment; and Sterilizer, Surgical Instrument and Dressing.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this document.
b. Issue of DODISS to be cited in the solicitation, and if required, the specific issue of individual documents referenced (see 2.1.1 and 2.2).
c. When a first article is required (see 3.1, 4.3, and 6.3).
d. Selection of applicable levels of preservation and packing (see 5.1 and 5.2).
e. When weather-resistant grade fiberboard shipping containers are required for level B packing (see 5.2.2.1).
f. When palletization is required (see 5.3).

6.3 First article. When a first article is required, it shall be inspected and approved under the appropriate provisions of FAR 52.209. The first article should be a preproduction sample. The contracting officer should specify the appropriate type of first article and the number of units to be furnished. The contracting officer should include specific instructions in all acquisition documents regarding arrangements for selection, inspection, and approval of the first article.

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6.4 Thread compound. Thread compound, Permatex No. 2 by Permatex Industrial Corp., 30 Tower Lane, Avon Park South, Avon, CT 06001 has been found to meet the requirements of MIL-S-7916.

6.5 Acceptance criteria. The acceptance criteria below is recommended for use. The acceptance criteria as specified in the contract or purchase order shall be binding. Unless otherwise specified, the following acceptance criteria are in accordance with MIL-STD-105.

6.5.1 Hardness testing. An acceptable quality level (AQL), expressed in terms of defects per hundred units, of 4.0 is recommended.

6.5.2 End item visual examination. An AQL, expressed in terms of defects per hundred units, of 2.5 for major defects and 6.5 for total (major and minor combined) defects is recommended.

6.5.3 End item dimensional examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.5.4 Packaging examination. An AQL, expressed in terms of defects per hundred units, of 2.5 is recommended.

6.5.5 Palletization examination. An AQL, expressed in terms of defects per hundred units, of 6.5 is recommended.

6.6 Subject term (key word) listing.

Burner
Cabinet
Field equipment
Gasoline burner
Oven
Sterilizer

6.7 Changes from previous issue. Marginal notations are not used in this revision to identify changes with respect to the previous issue due to the extensiveness of the changes.

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Custodians:

Army - GL
Navy - YD

Review activities:

Army - MD
DLA - GS

User activity:

Navy - MC

Preparing activity:

Army - GL

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